

Missingness-aware Federated Contrastive Learning on Semantic Graphs

Shuo Yu
Dalian University of Technology
Dalian, China
shuo.yu@ieee.org

Zhuoyang Han
Dalian University of Technology
Dalian, China
42217022@mail.dlut.edu.cn

Guoqing Han
Dalian University of Technology
Dalian, China
hanguoqing666@outlook.com

Tao Tang
Zhejiang University of Technology
Zhejiang, China
tao.tang@ieee.org

Feng Ding*
Dalian University of Technology
Dalian, China
dingfeng@dlut.edu.cn

Qiang Zhang*
Dalian University of Technology
Dalian, China
zhangq@dlut.edu.cn

Abstract

Semantic graphs are fundamental to the Web, enabling applications such as semantic search, recommendation, and knowledge-intensive reasoning. In decentralized Web environments, however, these graphs are distributed across organizations and constrained by strict privacy policies, making centralized training infeasible. Federated learning provides a promising solution, yet its effectiveness is severely limited by the *dual incompleteness* of real-world semantic graphs: missing node attributes and incomplete relational structures. Such dual missingness, often heterogeneous and unobserved across clients, causes substantial degradation in model performance. We present **FedCL**, a missingness-aware federated contrastive learning framework for dual-incomplete semantic graphs. FedCL introduces two key components: a topology estimation module, grounded in rate-distortion theory, that privately quantifies structural incompleteness across clients, and a federated reconstruction module that leverages these estimations to generate plausible relations without inferring sensitive attributes. To further improve robustness, FedCL integrates graph contrastive learning across reconstructed subgraphs, ensuring semantic consistency across heterogeneous and incomplete client graphs. Experiments on benchmark citation and Web datasets demonstrate that FedCL consistently outperforms state-of-the-art baselines in accuracy and robustness under heterogeneous missingness, while preserving strong privacy guarantees. These results highlight FedCL as a scalable and trustworthy approach for federated learning on incomplete semantic graphs, advancing privacy-preserving knowledge sharing on the Web.

CCS Concepts

• **Information systems** → **Social networks**; • **Computing methodologies** → **Distributed computing methodologies**; • **Security and privacy** → **Human and societal aspects of security and privacy**.

*Corresponding Authors.



This work is licensed under a Creative Commons Attribution 4.0 International License.
WWW '26, Dubai, United Arab Emirates
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792413>

Keywords

Federated Graph Learning; Semantic Graphs; Dual Incompleteness; Missingness-aware Learning; Graph Contrastive Learning

ACM Reference Format:

Shuo Yu, Zhuoyang Han, Guoqing Han, Tao Tang, Feng Ding, and Qiang Zhang. 2026. Missingness-aware Federated Contrastive Learning on Semantic Graphs. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3774904.3792413>

1 Introduction

Semantic graphs play a central role in Web applications and services [1]. They underpin tasks such as semantic search, recommendation, question answering, and large-scale reasoning over knowledge bases [2, 3]. Unlike simple homogeneous graphs, semantic graphs encode rich information by representing both entities and the types of relations among them [4], making them a cornerstone of knowledge-driven Web services. In practice, however, these graphs are typically created and maintained by different organizations, platforms, or devices. Due to privacy regulations (e.g., GDPR) and business confidentiality, directly sharing the underlying data is often infeasible [5]. Federated learning (FL) provides a promising alternative by enabling collaborative model training without exposing raw graph data [6, 7].

A major challenge in federated learning on semantic graphs is their frequent incompleteness [8]. On the attribute side, nodes such as entities or user profiles may contain missing or redacted information. On the structural side, relations are often incomplete due to partial crawling, platform boundaries, or noisy extraction. For instance, cross-platform semantic graphs may capture only a subset of entity links, while recommendation graphs in online services often reflect limited user-item interactions. This *dual incompleteness* severely hampers learning, and naïve federated approaches trained directly on such graphs typically experience substantial performance degradation [9–13]. Existing studies generally address a single aspect—either imputing missing attributes [14, 15] or recovering missing structures [16, 17]—but struggle when both attributes and structures are incomplete. Some recent efforts attempt joint completion [18], yet these approaches remain difficult to deploy in federated settings because (i) inferring sensitive attributes introduces privacy risks, and (ii) inter-client coordination is limited. This leads to our first challenge: **Federated semantic graph learning**

must address dual incompleteness (attributes and topology) in a privacy-preserving manner without explicitly inferring sensitive attributes.

A second challenge arises from the inherent heterogeneity of Web data. Semantic graphs across clients can vary significantly in size, relation coverage, and density [19, 20]. Unlike closed-world datasets, Web graphs are open-ended and continuously evolving. Consequently, the degree of missingness is not only uneven across clients but also often unknown. Attribute missingness may sometimes be apparent—for example, when explicit null values are used—but structural incompleteness is much harder to detect because only partial subgraphs are observable. One client’s graph may include rich node attributes but sparse links, whereas another may provide dense relations but heavily redacted attributes. As illustrated in Figure 1, the combined effect of attribute and structural missingness is more detrimental than either alone, especially under heterogeneous client distributions. This brings us to the second challenge: **Federated semantic graph learning must remain robust by privately estimating and adapting to client-specific, open-world levels of incompleteness.**

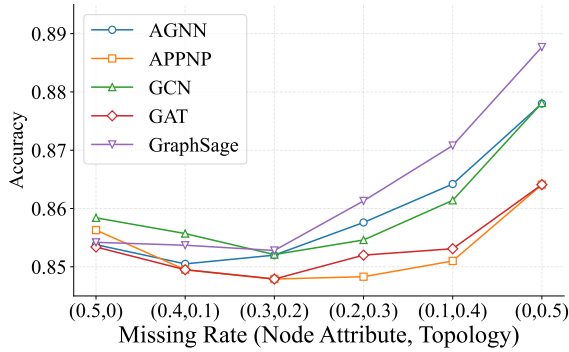


Figure 1: Performance on PubMed under varying node-attribute missing rates m^f and topology missing rates m^s ($m^f + m^s = 0.5$). Joint incompleteness amplifies performance degradation.

To tackle these challenges, we propose **FedCL**, a missingness-aware federated contrastive learning framework for semantic graphs. FedCL comprises two core modules: (i) a *missing topology estimation module*, grounded in rate-distortion theory, that quantifies structural information loss per client in a privacy-preserving manner; and (ii) a *topology reconstruction module* that stochastically generates plausible graph structures without exposing sensitive attributes. To enhance robustness, FedCL further integrates graph contrastive learning, enforcing embedding consistency across reconstructed graph views while preserving privacy. The resulting embeddings are aggregated via graph convolutional networks (GCNs) within the federated learning paradigm, enabling global training without raw data sharing.

Our main contributions are summarized as follows:

- We introduce **FedCL**, a federated learning framework for dual-incomplete semantic graphs that avoids risky attribute inference while achieving robust embedding alignment through contrastive learning.

- We design a *missing topology estimation module* based on rate-distortion theory, which privately quantifies heterogeneous structural incompleteness and guides adaptive reconstruction across clients.
- We conduct extensive experiments on benchmark citation and Web datasets, demonstrating that FedCL scales effectively to heterogeneous clients, integrates seamlessly with existing federated graph learning pipelines, and significantly mitigates performance degradation under dual incompleteness.

2 Methodology

2.1 Problem Definition

A global graph can be defined as $\mathcal{G} = \{V, E, X\}$, where V denotes the node set, X denotes the respective node attribute set, and E denotes the edge set. A subgraph federated learning system often contains one central server S and M clients with distributed subgraphs. $\mathcal{G}_i = \{V_i, E_i, X_i\}$ is the subgraph owned by client C_i , for $i \in [M]$, where $V = \bigcup_{i=1}^M V_i$. For simplicity, we assume that no overlapping nodes are shared across data owners, i.e., $V_i \cap V_j = \emptyset$ for any $i \neq j \in [M]$. For an edge $e_{v,u} \in E$, where $v \in V_i$ and $u \in V_j$, we have $e_{v,u} \notin E_i \cup E_j$, which means that $e_{v,u}$ is missing from the whole system but may exist in reality.

It is worth noting that the subgraphs from different client might be incomplete due to the data quality issues and privacy regulations, leading to missing attributes of nodes and edges. Compared with the complete subgraphs $\{\mathcal{G}_1, \dots, \mathcal{G}_M\}$, these incomplete subgraphs can be defined as $\{\bar{\mathcal{G}}_1, \dots, \bar{\mathcal{G}}_M\}$. For an incomplete subgraph $\{\bar{\mathcal{G}}_k\}$, the node set, edge set and attribute set are denoted as $\bar{V}_k, \bar{E}_k$ and $\bar{X}_k$ respectively, where $\bar{V}_k = V_k$.

2.2 Framework of FedCL

2.2.1 Overview. In this section, we introduce the proposed federated incomplete graph learning model, namely FedCL, whose overall structure is shown in Figure 2. FedCL consists of two core modules: the missing topology estimation module and the missing topology reconstruction module. Finally, within the federated learning framework, FedCL extends the above topology reconstruction mechanism to multi-client scenarios.

2.2.2 Missing Topology Estimation Module. We employ rate-distortion theory from information theory to quantify the impact of missing data on graph learning performance. Specifically, for a given graph $\mathcal{G} = \{V, E, X\}$, where V represents the node set, X denotes the respective node attribute set, and E represents the edge set, the proportion of missing node attributes is denoted as m^f ($0 \leq m^f \leq 1$). This means that, out of the total number of node attributes $|V| \times d$, a fraction m^f is missing, where d represents the dimensionality of node attributes. A particularly challenging scenario occurs when m^f of the nodes are missing all their attributes. Graphs with nodes missing all attributes are more difficult to train than those missing only a subset, which forms the basic scenario considered in our work.

From the perspective of rate-distortion theory, missing node attributes and missing topologies introduce information distortion into the graph. We assume that the total distortion is composed of two components: feature distortion $\mathcal{D}_f$ caused by missing node

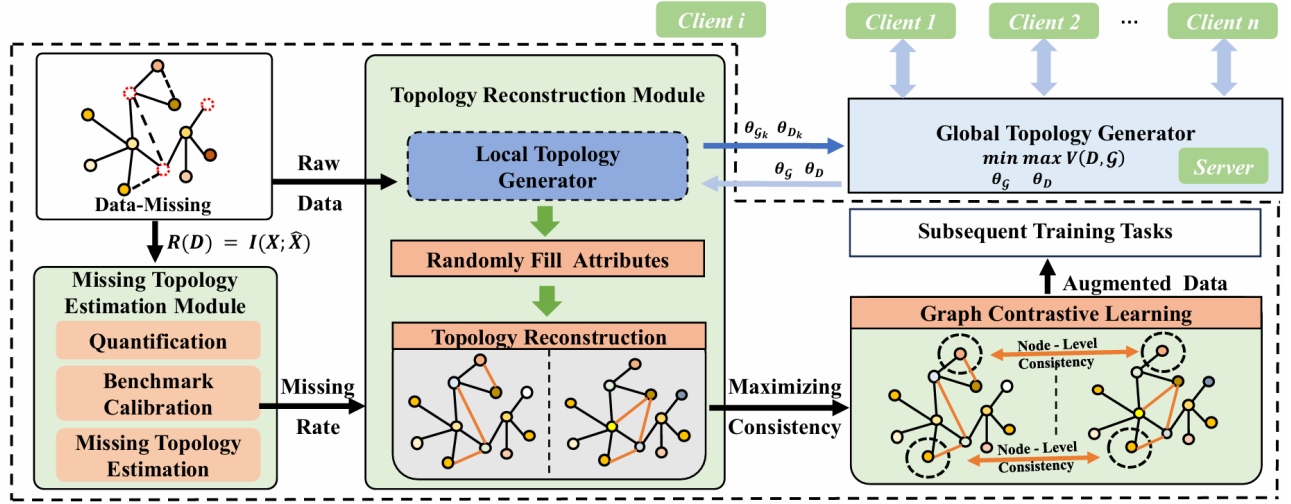


Figure 2: The framework of FedCL.

attributes and structural distortion $\mathcal{D}_s$ caused by missing edges. Specifically, we assume that the distortion is proportional to the missing rates:

$$\mathcal{D}_f = k_f m^f, \quad \mathcal{D}_s = k_s m^s, \quad (1)$$

where m^s denotes the topology missing rate, and k_f and k_s are proportionality constants characterizing the contribution of feature and structural missingness to information distortion, respectively.

According to rate-distortion theory, the effect of missing data on information distortion can be expressed as:

$$R(\mathcal{D}) = I(X; \hat{X}) = H(X) - H(X|\hat{X}), \quad (2)$$

where $\mathcal{D}$ represents distortion, $H(\cdot)$ denotes entropy, X represents the complete node attributes of $\mathcal{G}$, and $\hat{X}$ represents the node attributes of the incomplete graph. In graphs, X_i can be viewed as all the attributes of node i , and $I(X; \hat{X})$ can be interpreted as the mutual information between complete and missing node attributes. Obviously, at the input layer, we have $I(X_m; \hat{X}_m) = 0$, where $\hat{X}_m$ represents nodes with completely missing attributes.

In graph learning, the message-passing mechanism allows each node to aggregate information from its neighboring nodes and integrate it into its own attribute representation through an aggregation function. This mechanism mitigates the effect of missing node attributes to some extent by increasing the mutual information $I(X; \hat{X})$ for nodes with missing attributes. However, missing topologies weaken the message-passing mechanism's ability to enhance the mutual information $I(X; \hat{X})$. Therefore, both feature missingness and topology missingness jointly determine the effective distortion perceived by a graph neural network. In other words, during the first layer of a graph neural network, message passing enhances $I(X; \hat{X})$ for nodes with missing attributes, but its effectiveness depends on the underlying connectivity.

In classical communication systems, rate-distortion theory indicates that performance metrics such as bit error rate increase exponentially as the signal-to-noise ratio decreases, since reductions in mutual information have an exponential impact on task reliability.

Motivated by this principle, we assume that graph learning performance (e.g., node classification accuracy) decays exponentially with respect to the total distortion. In the following analysis, we use $ACC(m^f, m^s)$ to denote the performance of a graph learning model under given missing conditions. Specifically, $ACC(\cdot, \cdot)$ represents the evaluation metric of the downstream task (e.g., node classification accuracy) achieved when the feature missing rate is m^f and the topology missing rate is m^s . Although accuracy is adopted throughout this work for concreteness, the proposed formulation is not restricted to a specific metric and can be generalized to other performance measures such as F1-score or AUC. Accordingly, we model accuracy as:

$$ACC \propto e^{-c_f \mathcal{D}_f} \cdot e^{-c_s \mathcal{D}_s}, \quad (3)$$

where c_f and c_s are decay constants characterizing the impact strength of feature and structural distortion.

To further capture dataset- and model-specific characteristics, we introduce two sets of parameters. First, δ^f and δ^s denote dataset sensitivities to feature and structural missingness, respectively, reflecting how severely a dataset's performance degrades under distortion. Second, ϵ^f and ϵ^s represent model resistance to feature and structural missingness, respectively, characterizing the ability of a graph learning model (e.g., through aggregation and message passing) to mitigate distortion. Consequently, the net distortion effects can be expressed as $(\delta^f - \epsilon^f)m^f$ and $(\delta^s - \epsilon^s)m^s$.

Substituting these terms into the exponential performance decay model, we define the following quantification model:

$$ACC(m^f, m^s) = a \cdot e^{(\delta^f - \epsilon^f)m^f} \cdot e^{(\delta^s - \epsilon^s)m^s} + d, \quad (4)$$

where a is a scaling factor representing the baseline performance under complete data conditions, and d is a bias term that adjusts the accuracy offset. We set GCNs as the benchmark model, where $\epsilon_{GCN}^f = 0$ and $\epsilon_{GCN}^s = 0$, creating a neutral baseline in which performance variations are solely determined by dataset sensitivities. To calibrate δ^f and δ^s , we systematically train GCNs on each dataset under controlled combinations of node attribute missing rates m^f

and topology missing rates m^s , and fit the observed accuracies to:

$$ACC(m^f, m^s) = a \cdot e^{\delta^f m^f} \cdot e^{\delta^s m^s} + d. \quad (5)$$

This process yields dataset-specific sensitivity parameters, such as δ_{Cora}^f and δ_{Cora}^s . After calibration, the sensitivities are fixed, and the resistance parameters ϵ^f and ϵ^s of each candidate method are obtained by fitting their performance to the full quantification model in Equation (4).

This calibration process is repeated across all benchmark datasets to establish comprehensive sensitivity profiles. The attribute missing rate m^f is directly computable as $m^f = \frac{\text{count}(\text{missing attributes})}{|V| \times d}$, providing a known input parameter. By using the same experimental settings and publicly available benchmark datasets, the calibration process remains stable and reproducible. For new datasets, we can use publicly available benchmark datasets for parameter estimation to obtain dataset-specific sensitivity parameters. For the unknown topology missing rate m^s , we leverage pre-calibrated resistance parameters (ϵ^f , ϵ^s) from established models. The core procedure involves generating augmented dataset variants by introducing controlled additional missingness while preserving existing patterns. We then train models with known resistance parameters on these variants and record their performance metrics. Using the quantification model (Eq. 4) with fixed sensitivity parameters δ^f , δ^s and resistance parameters ϵ^f , ϵ^s , we formulate an inverse optimization problem where m^s becomes the primary variable. This nonlinear optimization minimizes the squared difference between observed accuracy and model-predicted accuracy across all augmented variants. To enhance robustness, the procedure is repeated with multiple pre-calibrated models, and final estimates are obtained through variance-weighted averaging of individual model solutions. This approach enables estimation of topology missing rates without prior knowledge of the dataset's complete structure.

2.2.3 Missing Topology Reconstruction Module. Reconstructing missing node attributes may lead to privacy leakage risks. Therefore, we only reconstruct the missing topologies by randomly filling in the missing node attributes instead of inferring them.

Specifically, we randomly delete a certain number of edges E_k^j from a local incomplete subgraph $\bar{G}_k$ to form a damaged subgraph $\mathcal{G}'_k$, where $E_k^j \in \bar{E}_k$. Hence, the damaged subgraph $\mathcal{G}'_k$ includes the complete node set $V'_k = V_k$, the node attributes $X'_k = \bar{X}_k$, and the edge set $E'_k = \bar{E}_k - E_k^j$. Notably, we assume that the two nodes connected by E_k^j have relatively complete node attributes.

Following GraphGAN [21], we train a topology generator $\mathcal{G}'_k$ and a topology discriminator $\mathcal{D}'_k$. We adopt the common adversarial mechanism in GAN networks using Equation (6).

$$\min_{\theta_G} \max_{\theta_D} V(D, G) = \sum_{c=1}^V \mathbb{E}_{v \sim p_{\text{true}}(\cdot|v_c)} [\log D(v, v_c; \theta_D)] + \mathbb{E}_{v \sim G(\cdot|v_c; \theta_G)} [\log(1 - D(v, v_c; \theta_D))], \quad (6)$$

where the generator G attempts to approximate $P_{\text{true}}(V|V_c)$ as closely as possible, finding nodes that are similar to the true neighbors of V_c to deceive the discriminator D . Meanwhile, the discriminator D determines whether a given node V is a true neighbor

of V_c or generated by G . In this way, the trained topology generator G_k can be used to generate additional edges to reconstruct the incomplete subgraph $\bar{G}_k$.

For an incomplete subgraph $\bar{G}_k$, two valid and distinct random values are used to fill the missing node attributes, resulting in two different attribute-repaired subgraphs, i.e., $\mathcal{G}_{k,a}$ and $\mathcal{G}_{k,b}$. These two subgraphs are then input into the topology generator $G_{\bar{G}_k}$ for topology reconstruction following Equation (6), producing the reconstructed subgraphs $\mathcal{G}'_{k,a}$ and $\mathcal{G}'_{k,b}$. It is worth noting that both reconstructed subgraphs are generated from the same incomplete subgraph $\bar{G}_k$ and are regarded as augmented views of the same original subgraph. Finally, a contrastive loss function is applied to maximize the consistency between the positive pairs of the two reconstructed subgraphs.

2.2.4 Federated Learning of FedCL. Our next goal is to deploy the proposed method within the federated learning framework and enable cross-client collaborative model training. In the federated setting, each client holds a local subgraph and performs local training independently. The global model is aggregated via the FedAvg algorithm to optimize node classification performance under dual missingness, without sharing raw graph data across clients.

In this setting, the incomplete subgraphs $\bar{G}_k$ within each client correspond to certain parts of the complete graph $\mathcal{G}$, with both missing node attributes and topologies. One challenge is that the performance of the client-side topology generator G_k and discriminator D_k may degrade due to insufficient data. Therefore, we apply FedAvg [22] to jointly train the generator and discriminator using all subgraphs from the clients:

$$\min_{\theta_{G_k}} \max_{\theta_{D_k}} V(D_k, G_k) = \sum_{c=1}^V \mathbb{E}_{v \sim p_{\text{true}}(\cdot|v_c)} [\log D_k(v, v_c; \theta_{D_k})] + \mathbb{E}_{v \sim G_k(\cdot|v_c; \theta_{G_k})} [\log(1 - D_k(v, v_c; \theta_{D_k}))], \quad (7)$$

$$\theta_G = \sum_{k=1}^M \frac{|V_k|}{\sum_{k \in M} |V_k|} \theta_{G_k}, \quad \theta_D = \sum_{k=1}^M \frac{|V_k|}{\sum_{k \in M} |V_k|} \theta_{D_k}. \quad (8)$$

Through federated learning, the graph contrastive learning can be further enhanced using the global G and D instead of the local G_k and D_k . Similarly, we also apply FedAvg to aggregate the node embeddings $\mathcal{H}_i$ of the incomplete subgraphs, which helps improve the performance of the local models.

Furthermore, FedCL is compatible with other advanced federated subgraph learning methods, such as FedSage [23], BDS-GCN [24], and FedGCN [25]. After the graph contrastive learning phase, the obtained node embeddings can be used for subsequent federated learning, enabling advanced graph federated learning methods to collaboratively train the global model.

2.2.5 Discussion. FedCL provides implicit privacy protection through local topology perturbation and restricted information sharing. Specifically, the random edge deletion operation introduces stochastic structural perturbations, which limits the influence of individual edges and provides a privacy effect conceptually similar to differential privacy. As a result, the existence of specific edges cannot be reliably inferred from the reconstructed topologies. Moreover,

within the federated learning framework, FedCL does not share raw node attributes or adjacency matrices across clients. Only topology-related model parameters are exchanged and aggregated via FedAvg, which encode high-level structural patterns rather than explicit local graph details. Since these parameters are averaged across multiple clients, the contribution of any single client is further diluted, thereby exposing limited topology information.

The missing topology estimation module solves a smooth single-variable inverse optimization problem, whose complexity is $O(\log(1/\epsilon))$ with respect to the optimization precision ϵ . The missing topology reconstruction module follows a GraphGAN-style training procedure, with a per-epoch computational complexity of $O(|E|d + |V|d^2)$, where $|E|$ and $|V|$ denote the numbers of edges and nodes, respectively, and d is the embedding dimension.

Before each round of training, we can pre-generate K reconstructed subgraphs and sequentially select two subgraphs for graph contrastive training in each iteration. It is important to note that these K reconstructed subgraphs are generated with different random values for filling the missing node attributes, which can be regarded as using different random seeds. When the number of selections reaches C_K^2 (i.e., all possible pairs are selected without repetition), we regenerate K new reconstructed subgraphs for training. This approach allows us to reduce the number of reconstructed subgraphs required for C_K^2 training rounds from $K(K-1)$ to just K . The value of K can be adjusted depending on the available storage space of the clients. Since the node identifiers in all reconstructed subgraphs of $\bar{\mathcal{G}}_k$ are consistent, it is feasible to employ binary encoding to merge and store the topologies. Specifically, the number of bits in the binary encoding represents the number of reconstructed subgraphs, and the value of each bit indicates the presence or absence of an edge. For instance, a given edge between nodes V_a and V_b has a value of 19; its binary representation is 10011, indicating the presence of the edge in the 1st, 2nd, and 5th reconstructed subgraphs, while absent in the others. This encoding strategy substantially reduces the storage requirements for reconstructed topologies, allowing all reconstructed subgraphs to be efficiently represented within a single unified adjacency matrix.

3 Experiments

3.1 Experimental Setup

3.1.1 Datasets. We conduct evaluations on nine benchmark datasets, as presented in Table 1. These datasets are categorized into three types based on their semantic properties. **Citation networks** (Cora, Citeseer, Pubmed, Citationv1, DBLPv7, ACMv9): These are domain-specific semantic graphs where nodes represent scholarly entities (papers or authors), edges encode relations like citations or topical relevance, and attributes correspond to textual features that capture research semantics. **Web networks** (Wikipedia, Email-Euore): Wikipedia consists of topic-related web pages with edges representing hyperlinks, while Email-Euore models organizational email communication networks, with attributes reflecting user profiles. **Social semantic graphs** (Polblogs): A network of political blogs where nodes represent blogs, edges indicate hyperlinks between blogs, and attributes encode textual content reflecting political stances (liberal/conservative). All datasets have been extensively used in prior works on graph representation learning and federated

graph learning, allowing us to comprehensively evaluate FedCL under varying degrees of attribute and topological incompleteness.

Table 1: The statistics of datasets.

Dataset	Nodes	Edges	Attributes	Classes
Wikipedia	2,405	11,596	2,405	17
Email-Euore	1,005	25,571	1,005	42
Cora	2,708	5,278	1,433	7
Citationv1	8,935	15,113	6,775	5
DBLPv7	5,484	8,117	6,775	5
ACMv9	9,360	15,556	6,775	5
Citeseer	3,312	4,732	3,703	6
Polblogs	1,224	16,718	1,224	2
Pubmed	19,717	44,338	500	3

Following the experimental setup of FedSage, we construct distributed subgraphs by partitioning each dataset among multiple participants, where each FL participant receives a subgraph corresponding to a part of the original graph. We select 5 clusters of approximately equal sizes to define subgraphs for the data owners, and allocate 60% of the data for training, 20% for validation, and the remaining 20% for testing.

3.1.2 Task Settings and Evaluation Metrics. The primary task of this work is node classification on semantic graphs under the federated learning paradigm, with a core focus on addressing graphs suffering from dual missingness (i.e., missing node attributes and incomplete topological edges). For the core node classification task, we use classification accuracy as the primary metric. Accuracy is defined as the ratio of correctly classified nodes to the total number of nodes in the test set, directly reflecting the model’s ability to learn discriminative node representations from incomplete graphs.

To validate the reliability of our proposed missingness quantification model, we employ four regression metrics:

- **Mean Squared Error (MSE):** Measures the average squared difference between predicted and actual values, quantifying overall prediction error.
- **Root Mean Squared Error (RMSE):** The square root of MSE, converting error to the original data scale for intuitive interpretation.
- **Mean Absolute Error (MAE):** Calculates the average absolute difference between predicted and actual values and is robust to outliers.
- **Coefficient of Determination (R^2):** Indicates how well the predicted values explain the variance in the actual data, reflecting model goodness-of-fit.

All metrics are computed on the test set of each client’s local subgraph, and results are averaged over 10 independent runs to ensure statistical stability.

3.1.3 Baselines. The baselines in this paper are categorized into three groups: evaluation models for node classification, generative-based graph learning approaches, and federated graph learning frameworks.

Table 2: Accuracy of node classification results on different graphs in the federated setting.

Method	Wikipedia	Email-Eucore	Cora	Citationv1	DBLPv7	ACMv9	Citeseer	Polblogs	Pubmed
SAT	0.6146	0.5598	0.7944	0.7607	0.7316	0.7303	0.6741	0.8513	0.8048
GraphMAE	0.6079	0.5531	0.7716	0.7484	0.7177	0.7045	0.6442	0.8484	0.7818
DGI	0.6045	0.5473	0.7813	0.7364	0.7212	0.7069	0.6469	0.8359	0.7824
GraphMAE2	0.6105	0.5563	0.7805	0.7486	0.7101	0.7232	0.6514	0.8371	0.7850
ASD-VAE	0.5905	0.5530	0.7963	0.7641	0.7382	0.7256	0.6654	0.8513	0.8121
AIAE	0.6013	0.5592	0.8018	0.7731	0.7296	0.7352	0.6583	0.8462	0.8079
FedCL	0.6224	0.5634	0.8042	0.7743	0.7441	0.7380	0.6691	0.8553	0.8152

The first category is used primarily for evaluation purposes and serves as standard benchmarks to assess node classification performance. These models represent well-established graph neural network architectures, including AGNN [26], APPNP [27], GAT [28], GCN, GraphSage [29], and SGC [30], providing a consistent basis for performance comparison across different settings. The second category consists of generative methods designed for incomplete graphs, including SAT [31], GraphMAE [32], DGI [33], GraphMAE2 [34], ASD-VAE [18], and AIAE [15]. These approaches that address missing attributes or structures through self-supervised learning, variational inference, or adversarial training techniques, frequently employing latent space modeling and attribute-structure decoupling strategie. The third category covers federated graph learning methods, such as FedSage [23], FedGCN [25], FedGraph [35], FED-PUB [36], and FedCog [37] that enable collaborative training on distributed subgraphs while preserving data privacy, utilizing mechanisms including neighbor generation, cross-client convolution, and personalized aggregation to overcome challenges in decentralized graph learning environments.

3.1.4 Implementation Details. All experiments are conducted on a workstation running Ubuntu 18.04.5 LTS, equipped with an Intel(R) Xeon(R) Silver 4114 CPU (2.20 GHz) and an NVIDIA Tesla V100S-PCIE GPU. Each reported result represents the average performance over ten independent runs to ensure statistical reliability. An early-stopping strategy with a patience of 500 is applied to prevent overfitting, and the dropout rate is fixed at 0.3. To obtain the resistance and sensitivity parameters, the learning rate is set to 0.003, with a hidden layer dimension of 64. Each model is trained for up to 10,000 epochs to ensure convergence. During topology reconstruction, FedCL performs dynamic gradient descent with a learning rate of 0.001 to update reconstruction parameters. In each iteration, the generator samples 20 instances, while the discriminator samples an equal number of positive instances corresponding to each vertex in the test set. Both the generator and discriminator are optimized alternately for 30 iterations per reconstruction cycle, and the embedding dimension is set to 20. For the contrastive learning component, we use the Adam optimizer with an initial learning rate of 0.001, the number of training epochs is fixed at 200, and the hidden dimension is 128.

3.2 Recommendation Performance

We compared FedCL with six representative generative-based graph learning methods under the FedAvg aggregation strategy, with both

node attribute and topology missing rates fixed at 0.4 to emulate real-world privacy-preserving collaboration scenarios.

As reported in Table 2, FedCL consistently outperforms all baseline methods on eight of the nine datasets. This improvement underscores its robustness across diverse graph structures and confirms the advantage of combining reconstruction-based data completion with federated contrastive learning. The Citeseer dataset has relatively sparse graph connectivity and lower homophily compared to other citation networks such as Cora and Pubmed. In such settings, neighbors often belong to different classes, making local message passing less reliable. FedCL’s topology reconstruction module, which assumes that missing edges can be inferred from structural similarity, may therefore struggle to generate useful connections when homophily is weak. The contrastive objective further amplifies this challenge, as it attempts to align global and local representations even when the underlying structural cues are noisy or ambiguous. These conditions can reduce the discriminative strength of learned embeddings relative to methods that rely purely on observed links (e.g., SAT).

3.3 Compatibility of FedCL in Existing Federated Graph Learning Frameworks

Table 3: Performance improvement of federated graph learning models integrated with FedCL

Dataset	FedSage	FedGCN	FedGraph	FED-PUB	FedCog
Wikipedia	+0.0071	+0.0046	+0.0113	+0.0093	+0.0124
Email-Eucore	+0.0153	+0.0136	+0.0043	+0.0063	+0.0108
Cora	+0.0012	+0.0023	+0.0026	+0.0018	-0.0029
Citationv1	-0.0030	+0.0101	+0.0052	+0.0045	+0.0027
DBLPv7	+0.0071	+0.0057	+0.0105	+0.0064	+0.0121
ACMv9	+0.0082	+0.0071	+0.0067	+0.0153	+0.0114
Citeseer	+0.0094	+0.0105	+0.0082	+0.0024	+0.0051
Polblogs	+0.0042	+0.0087	+0.0115	+0.0063	-0.0013
Pubmed	+0.0031	+0.0059	+0.0042	+0.0079	+0.0022

To assess FedCL’s compatibility across diverse federated graph learning architectures, we integrated its topology reconstruction and contrastive learning modules into five representative frameworks: FedSage, FedGCN, FedGraph, FED-PUB, and FedCog. We evaluated their performance under 40% attribute and topology missing rates on nine benchmark datasets.

As presented in Table 3, FedCL improves performance in 42 out of 45 model–dataset configurations, confirming its general effectiveness in mitigating the degradation caused by incomplete

graph data. The Cora dataset exhibits strong homophily, where most neighboring nodes belong to the same class, and the Polblogs network is small but densely connected. FedCog, which already employs cognitive attention for intra-community consistency, becomes susceptible to over-regularization when FedCL’s contrastive loss enforces additional similarity between positive pairs. This results in excessively smoothed node embeddings and reduced intra-class discriminability. Such representation collapse effects are consistent with prior contrastive graph studies, where strong regularization diminishes feature diversity on dense, homophilic graphs. FedSage performs localized aggregation inspired by GraphSAGE, using neighbor sampling and non-shared normalization across clients. When FedCL introduces global contrastive embeddings, discrepancies arise between global feature normalization and FedSage’s stochastic aggregation, leading to unstable gradient updates and minor accuracy degradation. Overall, FedCL enhances most models and datasets without altering their original architectures, and highlights its value as in privacy-preserving and incomplete data federated graph learning.

3.4 Evaluation of Missing Data Impact

We first assess the sensitivity of several popular GNN models to missing data, as well as the resistance of several classic datasets against missing data, as shown in Tables 4 and 5. Additionally, we calculate Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2) to evaluate the fit between our model predictions and actual data, thereby demonstrating how well our model reflects the sensitivity of GNNs and the resistance of datasets to missing data.

The Cora, PubMed, and Citeseer datasets exhibit high attribute sensitivity, indicating that they are particularly susceptible to missing node attributes. This suggests that the absence of node attributes significantly impacts model performance on these datasets. In contrast, DBLPv7 shows lower attribute sensitivity, indicating a reduced reliance on node attributes, as its topological information effectively compensates for missing attributes. Polblogs displays the highest topology sensitivity, highlighting its pronounced vulnerability to missing topologies. This suggests that models relying on this dataset heavily depend on topological information, consistent with its high average degree. This sensitivity may be attributed to the characteristics of social networks, where connections between nodes are crucial for the dissemination of content and influence. Conversely, DBLPv7 and Citeseer demonstrate lower topology sensitivity, indicating that these datasets maintain relatively robust performance even in the presence of missing edges.

GraphSAGE demonstrates high resistance to missing node attributes, reflecting its ability to effectively leverage topologies to compensate for missing attributes. In contrast, APPNP exhibits lower resistance to missing node attributes, suggesting that it relies heavily on node attributes for effective learning. Regarding missing topologies, APPNP, GAT, and GraphSAGE show relatively weak resistance, particularly in scenarios with substantial missing edges. In contrast, AGNN displays stronger resistance to missing topologies, likely due to its adaptive mechanism for updating node representations, which enhances its robustness to topological incompleteness.

SGC demonstrates notable advantages in resisting both missing node attributes and missing topologies. This may be attributed to its relatively lower performance on complete graphs, which suggests that it sacrifices optimal utilization of node attributes and topological information for computational simplification. Consequently, this characteristic can lead to unexpected robustness in the presence of missing data, allowing the model to maintain relatively high performance despite data incompleteness.

Table 4: Resistance of graph learning models to missing node attributes and topologies.

Model	Attribute	Topology	MSE↓	RMSE↓	MAE↓	R^2 ↑
GCN (Benchmark)	0	0	0.0005	0.0228	0.0164	0.9655
AGNN	0.0139	0.1276	0.0004	0.0205	0.0145	0.9650
APPNP	-0.1000	-0.2059	0.0004	0.0196	0.0154	0.9724
GAT	-0.0037	-0.1621	0.0004	0.0210	0.0154	0.9781
GraphSage	0.1895	-0.2192	0.0004	0.0197	0.0153	0.9745
SGC	0.2334	0.6484	0.0004	0.0200	0.0161	0.9653

Table 5: Sensitivity of datasets to missing node attributes and topologies.

Dataset	Attribute	Topology	MSE↓	RMSE↓	MAE↓	R^2 ↑
Wikipedia	0.7566	0.7924	0.0014	0.0372	0.0273	0.8918
Email-Eucore	1.2673	0.8370	0.0017	0.0388	0.0293	0.8858
Cora	1.6219	0.9387	0.0006	0.0236	0.0177	0.9734
Citationv1	0.9823	1.0494	0.0005	0.0225	0.0170	0.9693
DBLPv7	0.7571	0.7177	0.0005	0.0238	0.0174	0.9606
ACMv9	0.8582	0.8880	0.0008	0.0286	0.0214	0.9480
Citeseer	1.5397	0.7448	0.0002	0.0155	0.0118	0.9850
Polblogs	0.8561	1.4343	0.0002	0.0157	0.0114	0.9485
Pubmed	1.8159	1.1484	0.0001	0.0113	0.0087	0.9876

3.5 Missing Rate Evaluation Results

To provide a clearer interpretation of the evaluation outcomes, we visualize the correspondence between the empirical classification accuracy and the theoretically predicted accuracy derived from our quantification model. Representative examples are illustrated in Figure 3, Figure 4, and Figure 5. In each visualization, the response surface depicts the variation of node classification accuracy with respect to two independent variables: the attribute-missing proportion (X-axis) and the topology-missing proportion (Y-axis), both ranging from 0 to 0.9. The Z-axis represents the resulting accuracy value. For every experimental configuration, two corresponding surfaces are plotted: (1) the empirical surface, fitted directly from the observed evaluation results obtained under varying missing-attribute and missing-topology ratios, and (2) the theoretical surface, computed from the quantification model’s predicted accuracy values under the same parameter settings.

The predicted and empirical surfaces exhibit strong numerical and geometric correspondence, demonstrating that the quantification model accurately characterizes the interaction between missing attributes and incomplete topologies. These visual results confirm that the proposed quantification model reliably captures both the sensitivity of graph learning algorithms to missing information

and the resilience of different datasets to structural and attribute loss. Consequently, the model can be employed to estimate accuracy under arbitrary data-missing configurations, thereby guiding the selection and deployment of suitable federated graph learning strategies in privacy-constrained or incomplete-data scenarios.

In practical use, when sensitivity parameters for a new dataset are unavailable, practitioners can derive attribute-missing rates directly from feature null statistics and estimate topology-missing rates using the topology reconstruction module combined with baseline models. Conversely, the same quantification framework can serve as an analytical tool to benchmark novel methods by comparing their inferred parameters with canonical baselines, thus bridging theoretical performance modeling and empirical evaluation in federated graph learning.

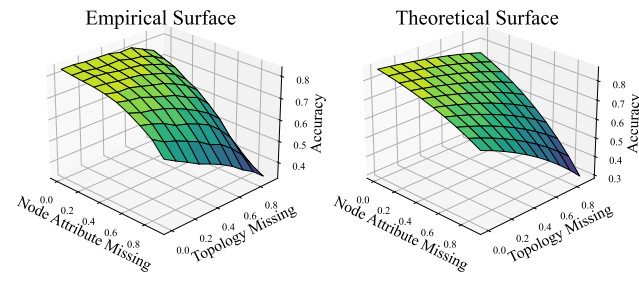


Figure 3: The the empirical surface and theoretical surface of APPNP across varying missing rates on ACMv9.

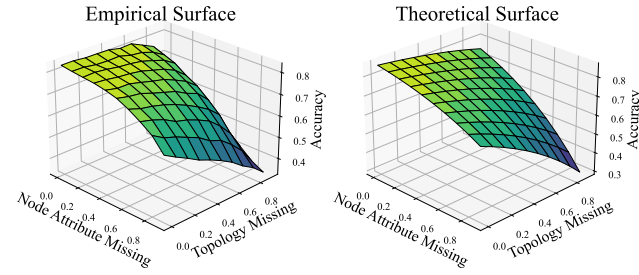


Figure 4: The the empirical surface and theoretical surface of GAT across varying missing rates on ACMv9.

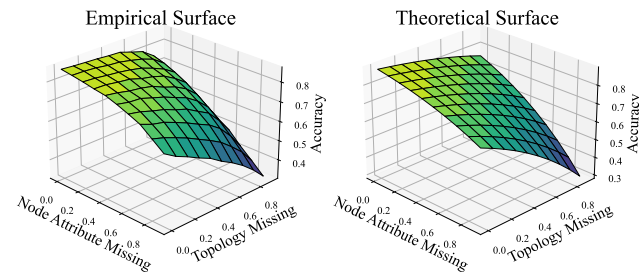


Figure 5: The the empirical surface and theoretical surface of GAT across varying missing rates on Citationv1.

3.6 Optimal Reconstruction Strategy

Table 6: Optimal reconstruction topology region for FedCL on Pubmed.

Missing Structure	Missing Node Attribute			
	0.1	0.2	0.3	0.4
0.1	[0.08,0.11]	[0.08,0.11]	[0.10,0.12]	[0.09,0.13]
0.2	[0.17,0.20]	[0.19,0.22]	[0.17,0.23]	[0.20,0.24]
0.3	[0.28,0.31]	[0.28,0.30]	[0.29,0.33]	[0.28,0.33]
0.4	[0.39,0.41]	[0.38,0.40]	[0.39,0.41]	[0.39,0.42]
0.5	[0.48,0.53]	[0.48,0.53]	[0.48,0.54]	[0.49,0.53]
0.6	[0.57,0.63]	[0.58,0.63]	[0.59,0.66]	[0.60,0.66]

Table 6 presents the reconstruction principles for the topology reconstruction process of FedCL on the PubMed dataset. Across a spectrum of structural losses (0.1–0.6) and node attribute missingness (0.1–0.4), the results reveal that the empirically derived optimal reconstruction regions exhibit strong consistency with the actual proportion of missing structural information. Specifically, the optimal reconstruction range is defined as the region in which the reconstructed topology yields accuracy values within 0.005 of the best-performing configuration, averaged over ten independent runs. A key observation is that the optimal reconstruction region maintains a tight alignment with the true structural loss ratio. Consequently, FedCL achieves stable recovery performance as long as the reconstruction parameter remains close to the actual topology deficit. Moreover, as structural loss increases from 0.1 to 0.6, the optimal reconstruction ranges expand slightly, but their central values consistently track the true missing ratio, demonstrating a near-linear correspondence. Such proportional adaptability ensures that reconstruction neither overfits the observed structure nor introduces excessive artificial edges, which could otherwise distort the graph’s semantic coherence. Therefore, we can use topology estimation module to estimate the proportion of missing topology, enabling us to identify the optimal reconstruction strategy.

4 Conclusion

In this paper, we addressed the critical challenge of dual incompleteness in semantic graphs within decentralized Web environments. We quantified the impact of missing data on graph learning models, and introduced FedCL to handle incomplete graphs, which reconstructs topology and applies graph contrastive learning, ensuring privacy preservation in federated settings. Experimental results demonstrate the effectiveness of FedCL. Future research will focus on improving the efficiency, scalability, and adaptability of federated graph learning systems, particularly under constrained computing resources and large-scale deployment scenarios. In addition, further examining the privacy–utility trade-off represent promising directions for advancing practical applications of federated graph learning in real-world incomplete-data environments.

Acknowledgments

This work is supported by the National Natural Science Foundation of China (No. 62576075).

References

- [1] Ben Liu, Jihai Zhang, Fangquan Lin, Cheng Yang, Min Peng, and Wotao Yin. Symagent: A neural-symbolic self-learning agent framework for complex reasoning over knowledge graphs. In *Proceedings of the ACM on Web Conference*, pages 98–108, 2025.
- [2] Zhenhao Xu, Mark Jerome Cruz, Matthew Guevara, Tie Wang, Manasi Deshpande, Xiaofeng Wang, and Zheng Li. Retrieval-augmented generation with knowledge graphs for customer service question answering. In *Proceedings of the 47th international ACM SIGIR conference on research and development in information retrieval*, pages 2905–2909, 2024.
- [3] Yangqin Jiang, Yuhao Yang, Lianghao Xia, and Chao Huang. Diffkg: Knowledge graph diffusion model for recommendation. In *Proceedings of the 17th ACM international conference on web search and data mining*, pages 313–321, 2024.
- [4] Xiao Long, Liansheng Zhuang, Aodi Li, Jiuchang Wei, Houqiang Li, and Shafei Wang. Kgdml: A diffusion model to capture multiple relation semantics for knowledge graph embedding. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 8850–8858, 2024.
- [5] Huan Wang, Haoran Li, Huaming Chen, Jun Yan, Jiahua Shi, and Jun Shen. Fedsc: Federated learning with semantic-aware collaboration. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2*, pages 2938–2949, 2025.
- [6] Lingyuan Meng, Ke Liang, Hao Yu, Yue Liu, Sihang Zhou, Meng Liu, and Xinwang Liu. Fedean: Entity-aware adversarial negative sampling for federated knowledge graph reasoning. *IEEE Transactions on Knowledge and Data Engineering*, 2024.
- [7] Zhuang Qi, Lei Meng, Zhaochuan Li, Han Hu, and Xiangxu Meng. Cross-silo feature space alignment for federated learning on clients with imbalanced data. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 19986–19994, 2025.
- [8] Yan Li, Xingchen Hu, Shengju Yu, Weiping Ding, Witold Pedrycz, Yeo Chai Kiat, and Zhong Liu. A vertical federated multi-view fuzzy clustering method for incomplete data. *IEEE Transactions on Fuzzy Systems*, 2025.
- [9] Linyu Li, Zhi Jin, Xuan Zhang, Haoran Duan, Jishu Wang, Zhengwei Tao, Haiyan Zhao, and Xiaofeng Zhu. Multi-view riemannian manifolds fusion enhancement for knowledge graph completion. *IEEE Transactions on Knowledge and Data Engineering*, 2025.
- [10] Bisheng Tang, Xiaojun Chen, Shaopu Wang, Yuxin Xuan, and Zhendong Zhao. Federated learning with limited node labels. *arXiv preprint arXiv:2406.12435*, 2024.
- [11] Yulan Hu, Sheng Ouyang, Zhirui Yang, Ge Chen, Junchen Wan, Xiao Wang, and Yong Liu. Exploring task unification in graph representation learning via generative approach. *arXiv preprint arXiv:2403.14340*, 2024.
- [12] Tao Tang, Zhuoyang Han, Zhen Cai, Shuo Yu, Xiaokang Zhou, Taiwo Oseni, and Sajal K. Das. Personalized federated graph learning on non-iiid electronic health records. *IEEE Transactions on Neural Networks and Learning Systems*, pages 1–14, 2024.
- [13] Kun Wu and Wendy Hui Wang. Verification of incomplete graph unlearning through adversarial perturbations. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2*, pages 3204–3215, 2025.
- [14] Liang Peng, Nan Wang, Nicha Dvornek, Xiaofeng Zhu, and Xiaoxiao Li. Fedni: Federated graph learning with network inpainting for population-based disease prediction. *IEEE Transactions on Medical Imaging*, 42(7):2032–2043, 2023.
- [15] Riting Xia, Chunxu Zhang, Anchen Li, Xueyan Liu, and Bo Yang. Attribute imputation autoencoders for attribute-missing graphs. *Knowledge-Based Systems*, 291:111583, 2024.
- [16] Ghazaleh Niknam, Soheila Molaei, Hadi Zare, David Clifton, and Shirui Pan. Graph representation learning based on deep generative gaussian mixture models. *Neurocomputing*, 523:157–169, 2023.
- [17] Xunkai Li, Zhengyu Wu, Wentao Zhang, Yinlin Zhu, Rong-Hua Li, and Guoren Wang. Fedgta: Topology-aware averaging for federated graph learning. *Proc. VLDB Endow.*, 17(1):41–50, September 2023.
- [18] Xinke Jiang, Zidi Qin, Jiarong Xu, and Xiang Ao. Incomplete graph learning via attribute-structure decoupled variational auto-encoder. In *Proceedings of the 17th ACM International Conference on Web Search and Data Mining*, pages 304–312, 2024.
- [19] Wenxuan Tu, Renxiang Guan, Sihang Zhou, Chuan Ma, Xin Peng, Zhiping Cai, Zhe Liu, Jieren Cheng, and Xinwang Liu. Attribute-missing graph clustering network. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(14):15392–15401, 2024.
- [20] Jia Wang, Yawen Li, Zhe Xue, Yingxia Shao, Zeli Guan, and Wenling Li. Horizontal federated heterogeneous graph learning: A multi-scale adaptive solution to data distribution challenges. In *Proceedings of the ACM on Web Conference 2025*, pages 4582–4591, 2025.
- [21] Hongwei Wang, Jia Wang, Jialin Wang, Miao Zhao, Weinan Zhang, Fuzheng Zhang, Xing Xie, and Minyi Guo. Graphgan: Graph representation learning with generative adversarial nets. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1), 2018.
- [22] Adrian Nilsson, Simon Smith, Gregor Ulm, Emil Gustavsson, and Mats Jirstrand. A performance evaluation of federated learning algorithms. In *Proceedings of the second workshop on Distributed Infrastructures for Deep Learning*, pages 1–8, 2018.
- [23] Ke Zhang, Carl Yang, Xiaoxiao Li, Lichao Sun, and Siu Ming Yiu. Subgraph federated learning with missing neighbor generation. In *Thirty-Fifth Conference on Neural Information Processing Systems*, 2021.
- [24] Cheng Wan, Youjie Li, Ang Li, Nam Sung Kim, and Yingyan Lin. BNS-GCN: Efficient full-graph training of graph convolutional networks with partition-parallelism and random boundary node sampling. *Proceedings of Machine Learning and Systems*, 4:673–693, 2022.
- [25] Yuhang Yao, Weizhao Jin, Srivatsan Ravi, and Carlee Joe-Wong. Fedgen: Convergence-communication tradeoffs in federated training of graph convolutional networks. In *Advances in Neural Information Processing Systems*, volume 36, pages 79748–79760, 2023.
- [26] Kiran K Thekumparampil, Chong Wang, Sewoong Oh, and Li-Jia Li. Attention-based graph neural network for semi-supervised learning. *arXiv preprint arXiv:1803.03735*, 2018.
- [27] Johannes Gasteiger, Aleksandar Bojchevski, and Stephan Günnemann. Predict then propagate: Graph neural networks meet personalized pagerank. *arXiv preprint arXiv:1810.05997*, 2018.
- [28] Petar Velickovic, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. Graph attention networks. In *Proceedings of the 6th International Conference on Learning Representations*, 2018.
- [29] Will Hamilton, Zhitaoying, and Jure Leskovec. Inductive representation learning on large graphs. *Advances in Neural Information Processing Systems*, 30, 2017.
- [30] Felix Wu, Amauri H. Souza Jr., Tianyi Zhang, Christopher Fifty, Tao Yu, and Kilian Q. Weinberger. Simplifying graph convolutional networks. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97, pages 6861–6871, 2019.
- [31] Xu Chen, Siheng Chen, Jiangchao Yao, Huangjie Zheng, Ya Zhang, and Ivor W Tsang. Learning on attribute-missing graphs. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(2):740–757, 2020.
- [32] Zhenyu Hou, Xiao Liu, Yukuo Cen, Yuxiao Dong, Hongxia Yang, Chunjie Wang, and Jie Tang. Graphmae: Self-supervised masked graph autoencoders. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 594–604, 2022.
- [33] Petar Velickovic, William Fedus, William L Hamilton, Pietro Liò, Yoshua Bengio, and R Devon Hjelm. Deep graph infomax. In *International Conference on Learning Representations*, 2019.
- [34] Zhenyu Hou, Yufei He, Yukuo Cen, Xiao Liu, Yuxiao Dong, Evgeny Kharlamov, and Jie Tang. Graphmae2: A decoding-enhanced masked self-supervised graph learner. In *Proceedings of the ACM Web Conference 2023*, pages 737–746, 2023.
- [35] Fahao Chen, Peng Li, Toshiaki Miyazaki, and Celimuge Wu. Fedgraph: Federated graph learning with intelligent sampling. *IEEE Transactions on Parallel and Distributed Systems*, 33(8):1775–1786, 2021.
- [36] Jinheon Baek, Wonyong Jeong, Jongdo Jin, Jaehong Yoon, and Sung Ju Hwang. Personalized subgraph federated learning. In *Proceedings of the 40th International Conference on Machine Learning*, pages 1396–1415, 2023.
- [37] Runze Lei, Pinghui Wang, Junzhou Zhao, Lin Lan, Jing Tao, Chao Deng, Junlan Feng, Xidian Wang, and Xiaohong Guan. Federated learning over coupled graphs. *IEEE Transactions on Parallel and Distributed Systems*, 34(4):1159–1172, 2023.
- [38] Xiaoye Miao, Yangyang Wu, Lu Chen, Yunjun Gao, and Jianwei Yin. An experimental survey of missing data imputation algorithms. *IEEE Transactions on Knowledge and Data Engineering*, 35(7):6630–6650, 2023.
- [39] Hanxuan Yang, Qingchao Kong, and Wenji Mao. Multi-level task-agnostic graph representation learning with isomorphic-consistent variational graph auto-encoders. *IEEE Transactions on Knowledge and Data Engineering*, 2025.
- [40] Lev Telyatnikov and Simone Scardapane. Egg-gae: scalable graph neural networks for tabular data imputation. In *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics*, volume 206, pages 2661–2676, 2023.
- [41] Jiajun Zhong, Ning Gui, and Weiwei Ye. Data imputation with iterative graph reconstruction. *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(9):11399–11407, 2023.
- [42] Ziqi Gao, Yifan Niu, Jiashun Cheng, Jianheng Tang, Lanqing Li, Tingyang Xu, Peilin Zhao, Fugee Tsung, and Jia Li. Handling missing data via max-entropy regularized graph autoencoder. *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(6):7651–7659, 2023.
- [43] Xindi Zheng, Yuwei Wu, Yu Pan, Wanyu Lin, Lei Ma, and Jianjun Zhao. Dpgan: A dual-path generative adversarial network for missing data imputation in graphs. *arXiv preprint arXiv:2404.17164*, 2024.
- [44] Mingliang Hou, Tao Tang, Feng Xia, Ibrahim Sultan, Roopdeep Kaur, and Xiangjie Kong. Missii: Missing information imputation for traffic data. *IEEE Transactions on Emerging Topics in Computing*, pages 1–14, 2023.
- [45] Zihan Tan, Guancheng Wan, Wenke Huang, He Li, Guibin Zhang, Carl Yang, and Mang Ye. Fedspa: Generalizable federated graph learning under homophily heterogeneity. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 15464–15475, 2025.
- [46] Ke Zhang, Lichao Sun, Bolin Ding, Siu Ming Yiu, and Carl Yang. Deep efficient private neighbor generation for subgraph federated learning. In *Proceedings of the 2024 SIAM International Conference on Data Mining (SDM)*, pages 806–814.

SIAM, 2024.

- [47] Riting Xia, Huibo Liu, Anchen Li, Xueyan Liu, Yan Zhang, Chunxu Zhang, and Bo Yang. Incomplete graph learning: A comprehensive survey. *Neural Networks*, page 107682, 2025.

A Related Work

A.1 Graph Learning on Attribute-Missing Data

Recent generative-based graph learning models for imputing missing node attributes can be broadly categorized into two paradigms: *graph auto-encoders (GAEs)* and *adversarial models* [38]. GAEs explicitly model the data generation process by learning priors or posteriors of node attributes, while adversarial models provide an implicit formulation by optimizing a log-likelihood surrogate rather than a direct reconstruction objective [39].

GAEs can be trained to recover missing node attributes while exploiting structural similarities during imputation, and have thus been widely adopted for handling incomplete graphs [40]. Zhong *et al.* [41] proposed an interactive graph generation and reconstruction framework to impute incomplete graphs. To alleviate spectral concentration issues in graph signals, Gao *et al.* [42] introduced a maximum-entropy regularized GAE that improves graph imputation by generating more diverse node attributes.

In parallel, generative adversarial networks (GANs) have also been widely used for attribute-missing graph learning. To address over-smoothing in graph propagation, Zheng *et al.* [43] proposed a dual-path GAN framework that jointly captures local and global representations of input graph signals to enhance long-range dependency modeling. The generator is trained with a local subgraph fidelity-based discriminator, which improves local imputation accuracy. Similarly, Hou *et al.* [44] proposed a theory-guided traffic graph GAN that incorporates real-world social information and human–social interactions based on spatial interaction theory.

However, these approaches inevitably involve explicit inference of missing node attributes, potentially leading to privacy leakage—particularly when attributes are intentionally withheld by users. This makes their direct application to federated learning scenarios challenging.

A.2 Federated Graph Learning

Federated graph learning (FGL) extends centralized graph learning to distributed environments, enabling joint training of a global model across multiple clients without sharing raw data [45]. FGL preserves data privacy while leveraging the structural and semantic information distributed across clients. Existing studies primarily focus on scenarios where client graphs are complete and belong to the same domain. For example, Zhang *et al.* [23] proposed a subgraph-based federated learning model that trains a global model using local subgraphs from each client. Missing links are estimated using node embeddings derived from a linear generative model, thereby improving efficiency.

To reduce communication overhead, Yao *et al.* [25] proposed a federated graph convolutional network (FedGCN) for semi-supervised node classification, incorporating homomorphic encryption to ensure privacy. When a node has only one neighbor stored on another client, FedGCN either removes that neighbor or protects it via differential privacy. To address missing cross-subgraph neighbors, Zhang *et al.* [46] introduced FedDEP, which employs a deep neighbor generator leveraging GNN embeddings of potential missing nodes and a pseudo-federated learning strategy for efficient neighbor synthesis through embedding prototyping.

Nevertheless, most current FGL methods remain ill-suited for incomplete graphs [47]. The number of reconstructed nodes is often smaller than that of the true missing ones, resulting in suboptimal imputations and accumulated errors, which degrade overall model performance.